

8 Hyperbolic functions	8.1	Understand the definitions of hyperbolic functions $\sinh x$, $\cosh x$ and $\tanh x$, including their domains and ranges, and be able to sketch their graphs.	For example, $\cosh x = \frac{1}{2}(e^x + e^{-x})$
	8.2	Differentiate and integrate hyperbolic functions.	For example, differentiate $\tanh 3x$, $x \sinh^2 x$, $\frac{\cosh 2x}{\sqrt{x+1}}$

Topic	What students need to learn:		
	Content		Guidance
8 Hyperbolic functions <i>continued</i>	8.3	Understand and be able to use the definitions of the inverse hyperbolic functions and their domains and ranges.	$\operatorname{arsinh} x = \ln \left[x + \sqrt{x^2 + 1} \right]$ $\operatorname{arcosh} x = \ln \left[x + \sqrt{x^2 - 1} \right], \quad x \geq 1$ $\operatorname{artanh} x = \frac{1}{2} \ln \left[\frac{1+x}{1-x} \right], \quad -1 < x < 1$
	8.4	Derive and use the logarithmic forms of the inverse hyperbolic functions.	
	8.5	Integrate functions of the form $(x^2 + a^2)^{-\frac{1}{2}}$ and $(x^2 - a^2)^{-\frac{1}{2}}$ and be able to choose substitutions to integrate associated functions.	